

AI HEALTHCARE ANALYTICS AGENT - USER GUIDE

- Developed By Ekta Grewal

Automated Patient & Healthcare Operations Insights System

1. Introduction

Project Overview

The AI Healthcare Analytics Agent is an automated healthcare operations analytics solution built using n8n, Microsoft SQL Server, Google Gemini, JavaScript, n8n Forms, and Gmail. The system converts natural-language healthcare questions into SQL queries, retrieves data from the healthcare database, calculates operational KPIs, analyzes monthly trends, detects unusual patient-volume patterns, generates AI-powered healthcare insights, and delivers the final analysis through Gmail.

Objective

To automate healthcare operational monitoring and provide clear, data-driven insights without requiring manual SQL analysis or manual report preparation.

2. System Architecture

Components Used

- n8n Local – Workflow automation and orchestration platform
- Microsoft SQL Server – Healthcare data storage
- Google Gemini – AI query planning and healthcare insight generation
- JavaScript – KPI, trend and anomaly calculations
- n8n Form – Natural-language healthcare question input
- Gmail – Automated healthcare report distribution

Workflow Overview

- User submits a healthcare analytics request through the n8n form.
- AI Query Planner converts the request into one Microsoft SQL Server query.
- SQL Server executes the generated query against the healthcare database.
- Healthcare KPI Calculation processes and summarizes the returned data.
- Trend Analysis calculates monthly patient-volume trends.
- Anomaly Detection compares monthly volume against the historical baseline.
- Operational Analysis combines the calculated healthcare analysis.
- Healthcare Insight Agent generates a clear final answer.
- Gmail sends the healthcare analysis automatically.

3. Data Sources

Healthcare Database Tables

- **Patients** – Patient demographic and insurance information.
- **Departments** – Department names, locations, capacity and department heads.
- **Admissions** – Patient admissions, dates, admission type, length of stay, readmission and status.
- **Billing** – Hospital charges, insurance coverage, patient responsibility and billing status.
- **Diagnoses** – Diagnosis codes, names, categories and primary diagnosis information.
- **Medications** – Medication usage, quantities, categories and medication costs.
- **Procedures** – Procedures, categories and procedure costs.
- **Mortality** – Mortality records, dates of death and cause categories.

Data Relationships

- Patients → Admissions
- Departments → Admissions
- Admissions → Billing
- Admissions → Diagnoses
- Admissions → Medications
- Admissions → Procedures
- Admissions → Mortality

The related healthcare tables are used to answer operational questions while avoiding incorrect totals caused by directly multiplying multiple one-to-many relationships.

4. Accessing the System

Healthcare Web Form

Users access the solution through the local n8n form. The user enters a healthcare question in natural language and submits the form.

Example Questions

- Give me a complete healthcare operational summary.
- Which department had the highest patient volume?
- Which month had the highest patient volume?
- Show the most frequently used medicines.
- Show the most common diagnoses.
- Which month had the highest hospital charges?

After successful submission:

- The healthcare database is queried automatically.
- KPIs and trends are calculated.
- Anomaly analysis is performed.
- An AI-generated healthcare answer is produced.
- The final answer is sent through Gmail.

5. Healthcare KPI Metrics Generated

The system can automatically calculate or return operational metrics such as:

- Patient Volume
- Admission Count
- Readmission Count
- Average Length of Stay
- Total Hospital Charges
- Medication Usage
- Procedure Count
- Mortality Count
- Department Patient Volume
- Monthly Patient Volume
- Month-over-Month Patient Volume Change

6. Healthcare Operational Intelligence

The system can generate insights across several operational areas.

Patient & Department Intelligence

- Highest patient-volume department
- Department-wise patient volume
- Patient-volume comparisons

Trend Intelligence

- Monthly patient-volume trends
- Highest-volume month
- Lowest-volume month
- Month-over-month percentage changes

Financial Intelligence

- Monthly hospital-charge analysis
- Daily hospital-charge analysis associated with admissions

Clinical Operations Intelligence

- Medication usage patterns
- Most frequently used medicines
- Most common diagnoses
- Procedure activity
- Mortality counts

7. Anomaly Detection

The current workflow uses monthly patient volume as the primary anomaly signal.

- A historical monthly baseline is calculated.
- Average monthly patient volume is calculated.
- Lower and upper thresholds are calculated using the historical variation.
- Each month is compared with the calculated thresholds.
- Months outside the expected range are flagged as anomalies.

Historical Trend Analysis

- All available months are returned chronologically for trend questions.
- Month-over-month patient-volume percentage changes are calculated.
- Highest and lowest patient-volume months are identified.
- Historical comparison is used when enough monthly data is available.

8. AI Healthcare Analysis

Google Gemini generates the final natural-language healthcare analysis using only the calculated workflow results.

Final AI Response

- Answers the user's actual healthcare question first.
- Uses the values returned by the SQL and calculation stages.
- Explains trends and operational patterns in simple language.
- Highlights relevant anomalies when present.
- Avoids inventing numbers or unsupported information.
- Does not provide medical diagnosis or medical advice.
- Clearly distinguishes data patterns from causation.

9. Generated Outputs

Healthcare Analytics Result

The main output is a natural-language healthcare analytics answer generated by the Healthcare Insight Agent.

Gmail Report

- The final healthcare answer is automatically sent through Gmail.
- The email can contain the formatted healthcare analysis and key operational insights.

Example Tested Outputs

- Highest patient-volume department: Emergency — 1,976 patients.
- Highest monthly patient volume: December 2025 — 556 patients.
- Lowest monthly patient volume: June 2024 — 415 patients.
- Daily operational report date: December 31, 2025.
- Daily patient volume: 27.
- Daily admission count: 27.
- Daily readmission count: 4.
- Daily average length of stay: 5.63 days.
- Daily hospital charges: \$220,780.82.
- Daily medication usage: 43.
- Daily procedure count: 23.
- Daily mortality count: 1.

Important

These are examples from the current healthcare dataset and tested workflow results; actual outputs change according to the question and current database data.

10. Troubleshooting

- No data returned: verify that the generated SQL is valid and that the healthcare tables contain data.
- Unexpected result: verify the selected table relationships and aggregation logic.
- AI answer missing: verify that the Healthcare Insight Agent has its Gemini model connected.
- Email not received: check the Gmail credential, recipient address and spam/junk folder.
- Form response missing: verify that Form Ending still references the Healthcare Insight Agent output.

11. Project Deliverables

- Natural-language healthcare query interface
- AI SQL query planning
- Microsoft SQL Server healthcare database integration
- Automated healthcare KPI computation
- Monthly trend analysis
- Patient-volume anomaly detection
- Department performance analysis
- Medication and diagnosis intelligence
- AI-generated healthcare operational insights
- Automated Gmail reporting

12. End-to-End Workflow

On Form Submission → AI Query Planner → Microsoft SQL → Healthcare KPI Calculation → Trend Analysis SQL → Trend Analysis → Anomaly Detection → Operational Analysis → Healthcare Insight Agent → Format Email → Gmail → Form Ending

13. Conclusion

The AI Healthcare Analytics Agent automates the healthcare operational analytics lifecycle from natural-language questions and SQL-based data retrieval to KPI computation, trend analysis, anomaly detection, AI-generated insights and automated email reporting.

The solution is designed to reduce manual analysis effort, improve visibility into patient and operational trends, and help healthcare teams access data-driven operational information faster.

Project Scope Note

This project is focused on healthcare operations and analytics. It is not intended to provide medical diagnosis, treatment recommendations, or clinical decision-making.

End of User Guide